

- 1 Onions are vegetables that are harvested from the plant *Allium cepa*. In many countries, small onions are used to make pickled onions. Pickled onions can be stored for a long time because the growth of microorganisms is prevented.

Pickled onions are made using this method.

- Remove the dry outer layer from small onions.
- Place the onions in a sodium chloride solution for several hours.
- Remove the sodium chloride solution and place the onions in a solution of ethanoic acid (vinegar).

Fig. 1.1 shows a container of pickled onions and some fresh onions that have **not** been pickled.



Fig. 1.1

- (a) Some students decided to investigate how placing small onions in different concentrations of sodium chloride solution affects the mass of the onions.
- (i) Identify the **independent** variable in this investigation.

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- (ii) The students were given a 20.0% stock solution of sodium chloride. The students decided to use the stock solution to make sodium chloride solutions with concentrations of 0.0%, 1.0%, 5.0%, 10.0%, 15.0% and 20.0%.

The students made 140 cm^3 of these solutions in separate beakers.

Complete Table 1.1 to show how **140 cm^3** of these solutions could be made by **proportional** dilution of the 20.0% stock solution of sodium chloride.

Table 1.1

percentage concentration of sodium chloride solution	volume of 20.0% sodium chloride solution / cm^3	volume of / cm^3
0.0		
1.0		
5.0		
10.0		
15.0		
20.0		

[2]

(b) The students carried out this procedure.

- The dry outer layer was removed from 30 small onions.
- Five of the small onions were placed into each of the six beakers containing the sodium chloride solutions prepared by the students.
- The onions were left in the sodium chloride solutions for **2 hours**.
- The mean percentage change in mass of the five small onions in each beaker was calculated.

The students then repeated the whole procedure using a new set of 30 small onions. This time the onions were left in the sodium chloride solutions for **48 hours**.

Fig. 1.2 shows the results of the investigation.

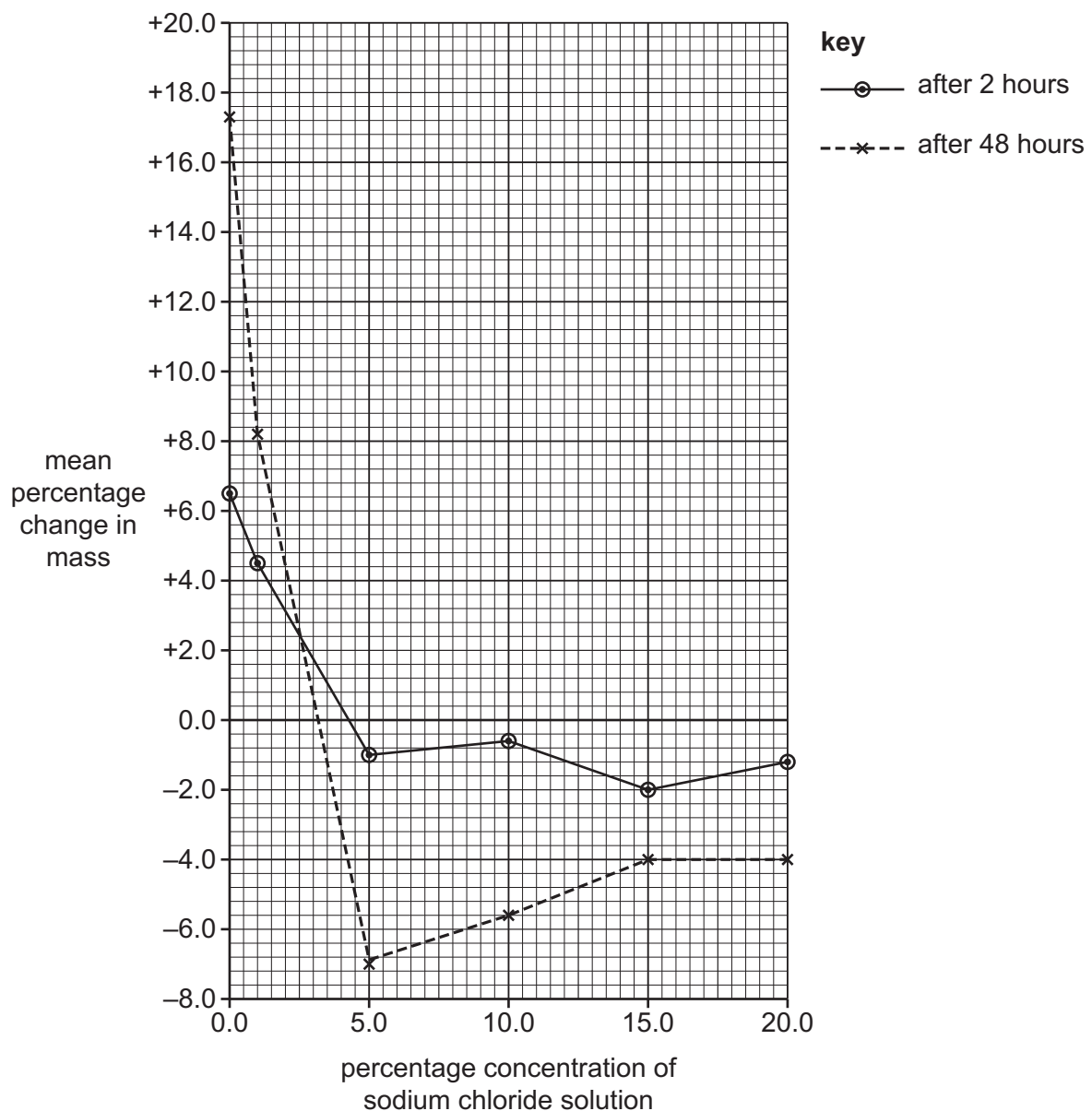


Fig. 1.2

- (i) Explain why the students calculated the percentage change in mass of the onions.

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..... [1]

- (ii) One of the students concluded that:

The water potential of the onion cells is the same as the water potential of a 4.2% sodium chloride solution.

With reference to the information provided, including the results shown in Fig. 1.2, suggest reasons why this conclusion should **not** be accepted.

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..... [4]

(c) The turnip, *Brassica rapa*, is a root vegetable. Turnips are grown in many parts of the world.

Fig. 1.3 shows freshly harvested turnips with the leaves still attached.

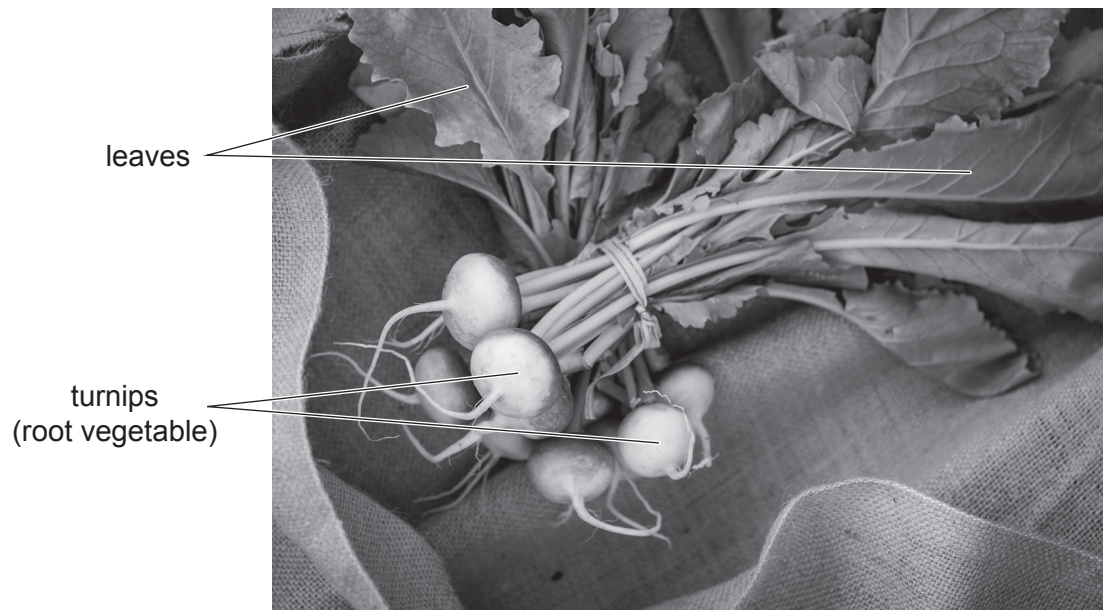


Fig. 1.3

A student decided to investigate the effect of temperature on the rate of osmosis in turnips.

In an initial test, the student removed the outer layer from a turnip and cut the turnip into small blocks. The student placed one of the turnip blocks into a beaker of distilled water, as shown in Fig. 1.4. Water entered the block by osmosis.

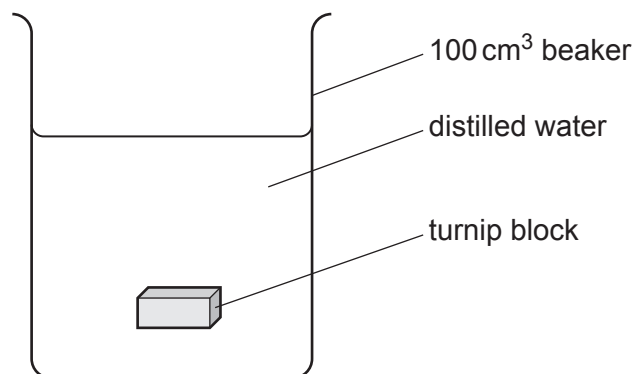


Fig. 1.4

- (ii) Complete the sketch graph shown in Fig. 1.5 to predict the effect of increasing the temperature, over a range from 10°C to 50°C, on the rate of osmosis in turnips.

Include axis labels with units in your answer.



Fig. 1.5

[2]

- (iii) Suggest **one** hazard of your method in (c)(i), the risk associated with the hazard and the precaution that would need to be taken.

hazard

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risk

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precaution

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[1]

[Total: 17]